

```
#
```

```
=====
```

```
==
```

```
# Roll no : 60    PRN: 0124UITM1060
```

```
# Name : Kshitij Shinde
```

```
# Dept. : Information Technology (Third Year)
```

```
#
```

```
=====
```

```
==
```

```
# Assignment 2:
```

```
# House Price Prediction using Linear Regression
```

```
#
```

```
=====
```

```
==
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
```

```
# Load Dataset
```

```
df = pd.read_csv("AMLIS/house_price.csv")
```

```
# Display First Five Records
```

```
print("First Five Records:")
```

```
print(df.head())
```

```
# Dataset Information
```

```
print("\nDataset Information:")
```

```
print(df.info())
```

```
# Check Missing Values
```

```
print("\nMissing Values:")
```

```
print(df.isnull().sum())
```

```
# Features (Independent Variables)
```

```
X = df[['area', 'bedrooms', 'bathrooms', 'age','parking']]
```

```
# Target Variable (Dependent Variable)
```

```
y = df['price']
```

```
# Split Dataset into Training and Testing Sets
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
# Create Linear Regression Model
```

```
model = LinearRegression()
```

```
# Train the Model
```

```
model.fit(X_train, y_train)
```

```
# Predict House Prices
```

```
y_pred = model.predict(X_test)
```

```
# Display Predictions
```

```
print("\nActual Price\tPredicted Price")
```

```
for actual, predicted in zip(y_test.values, y_pred):
```

```
    print(f"{actual}\t\t{predicted:.2f}")
```

```
# Evaluation Metrics
```

```
mae = mean_absolute_error(y_test, y_pred)
```

```
mse = mean_squared_error(y_test, y_pred)
```

```
rmse = np.sqrt(mse)
```

```
r2 = r2_score(y_test, y_pred)
```

```
print("\nModel Evaluation")
```

```
print("MAE :", mae)
```

```
print("MSE :", mse)
```

```
print("RMSE :", rmse)
```

```
print("R2 Score :", r2)
```

```
# Scatter Plot (Actual vs Predicted)

plt.figure(figsize=(6, 5))

plt.scatter(y_test, y_pred, color='blue')

plt.xlabel("Actual Price")

plt.ylabel("Predicted Price")

plt.title("Actual vs Predicted House Prices")

plt.grid(True)

plt.show()
```

#output

First Five Records:

	price	area	bedrooms	bathrooms	age	parking
0	13300000	7420	4	2	3	2
1	12250000	8960	4	4	4	3
2	12250000	9960	3	2	2	2
3	12215000	7500	4	2	2	3
4	11410000	7420	4	1	2	2

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 545 entries, 0 to 544

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
---	--------	----------------	-------

---	-----	-----	-----
-----	-------	-------	-------

0	price	545 non-null	int64
---	-------	--------------	-------

1	area	545 non-null	int64
---	------	--------------	-------

2	bedrooms	545 non-null	int64
---	----------	--------------	-------

3	bathrooms	545 non-null	int64
---	-----------	--------------	-------

4	age	545 non-null	int64
---	-----	--------------	-------

5	parking	545 non-null	int64
---	---------	--------------	-------

dtypes: int64(6)

memory usage: 25.7 KB

None

Missing Values:

price 0

area 0

bedrooms 0

bathrooms 0

age 0

parking 0

dtype: int64

Actual Price	Predicted Price
4060000	6178627.50
6650000	6370140.86
3710000	3283148.16
6440000	4226007.95
2800000	3409685.55
4900000	4262158.35
5250000	5493440.53
4543000	5559897.74
2450000	3373715.42
3353000	3020513.19
10150000	9519985.01
2660000	3250408.26
3360000	3184310.73

3360000	3639160.73
2275000	3907874.98
2660000	6174323.65
2660000	3148160.33
7350000	4872535.73
2940000	4826205.69
2870000	3888527.39
6720000	5222386.81
5425000	4294589.38
1890000	3162381.18
5250000	4193028.43
4193000	5747707.28
12250000	7619041.42
3080000	2989724.55
5110000	4794005.75
9800000	7434038.65
2520000	3457027.29
6790000	5102872.71
3500000	3982998.90

6650000	6215707.38
2940000	4724828.08
3325000	4307759.63
4200000	5625853.71
4900000	4791990.71
3290000	3815116.84
3500000	3261429.41
2380000	4879463.84
5495000	5326652.38
3675000	3156512.71
6650000	6345431.50
4907000	4316575.21
3150000	4120177.61
4480000	4190668.52
6580000	6710808.14
5740000	4603723.42
3003000	4521704.65
1820000	3637248.18
8400000	7841708.56

2450000	3299587.35
4270000	4772882.73
4007500	4071574.47
3234000	3957195.64
1750000	3085375.25
9800000	6971252.23
2100000	3422953.86
4340000	4825111.79
3045000	3209020.09
3850000	4878575.95
3500000	4526835.83
4753000	5037411.34
3080000	3567305.76
4550000	5012521.71
6510000	4518334.58
6685000	6744881.57
5110000	3608274.04
4550000	7430586.37
6650000	4622435.72

3640000	4747675.71
5600000	5141167.77
3780000	4967468.92
6615000	6953067.99
3220000	3201985.39
6650000	5334424.86
4690000	4478181.88
4830000	3522520.05
6860000	4377434.97
2233000	3823567.27
7035000	7153467.90
4165000	4617172.01
6195000	5746394.04
6510000	5236916.53
1890000	3206746.99
8190000	7231644.48
2660000	3051399.88
4193000	4084925.00
13300000	7480866.87

9681000	7732906.74
4340000	3572558.22
8645000	5010612.23
3703000	3718650.57
5145000	3734909.49
6440000	8681030.24
5950000	3975727.66
5810000	5623198.06
5740000	6649034.75
6230000	5213203.02
5600000	5562072.78
3010000	4357723.77
8890000	5963514.54
4900000	4094108.79
5530000	6072397.55
9100000	5361443.32
3773000	5311028.75
7910000	7199715.72
7350000	6058267.45

5530000 5931714.21

Model Evaluation

MAE : 1127483.3523235186

MSE : 2292721545725.3613

RMSE : 1514173.552049223

R2 Score : 0.5464062355495873

